JOHNS HOPKINS
WHITING SCHOOL of ENGINEERING

EN.520.681 Perception for Embodied AI

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Department of Electrical and Computer Engineering
Spring 2026

LABORATORY FOR Computational Sensing + Robotics
Data Science and AI Institute
JHU Electrical and Computer Engineering

General Information

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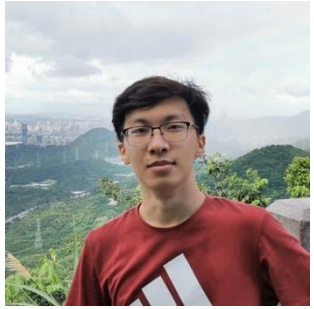
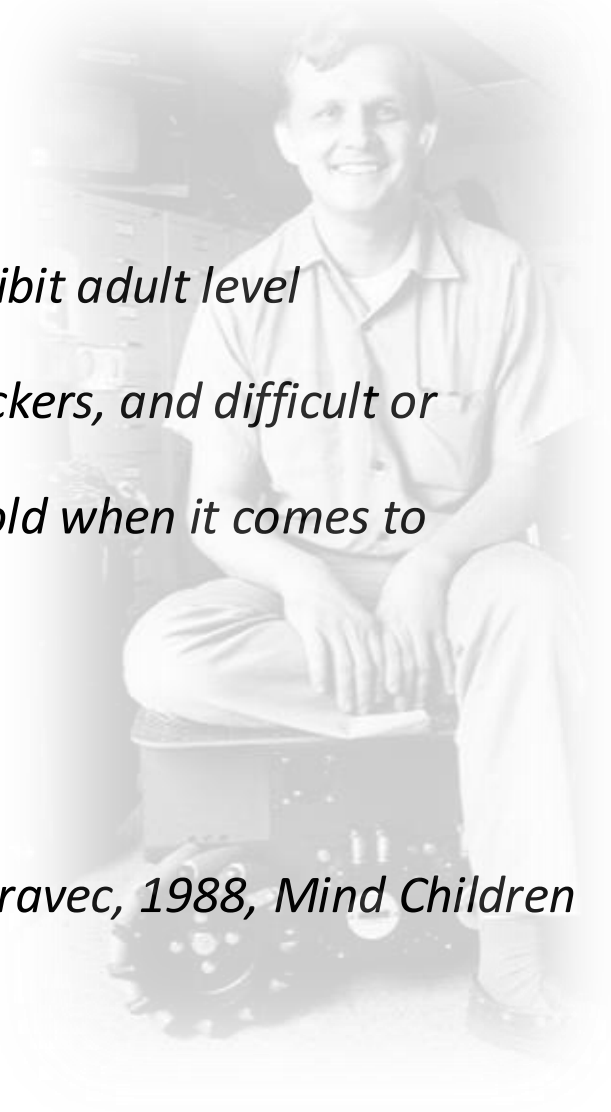
    - Office hours: Tuesday 9 – 10 am, Wednesday 4 – 5 pm, Barton 117



Today's Agenda

- Introduction
- Course Logistics
- Coordinate Systems, Rigid Transforms
- 3D Rotations

Moravec's Paradox



"It is comparatively easy to make computers exhibit adult level performance on intelligence tests or playing checkers, and difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility."

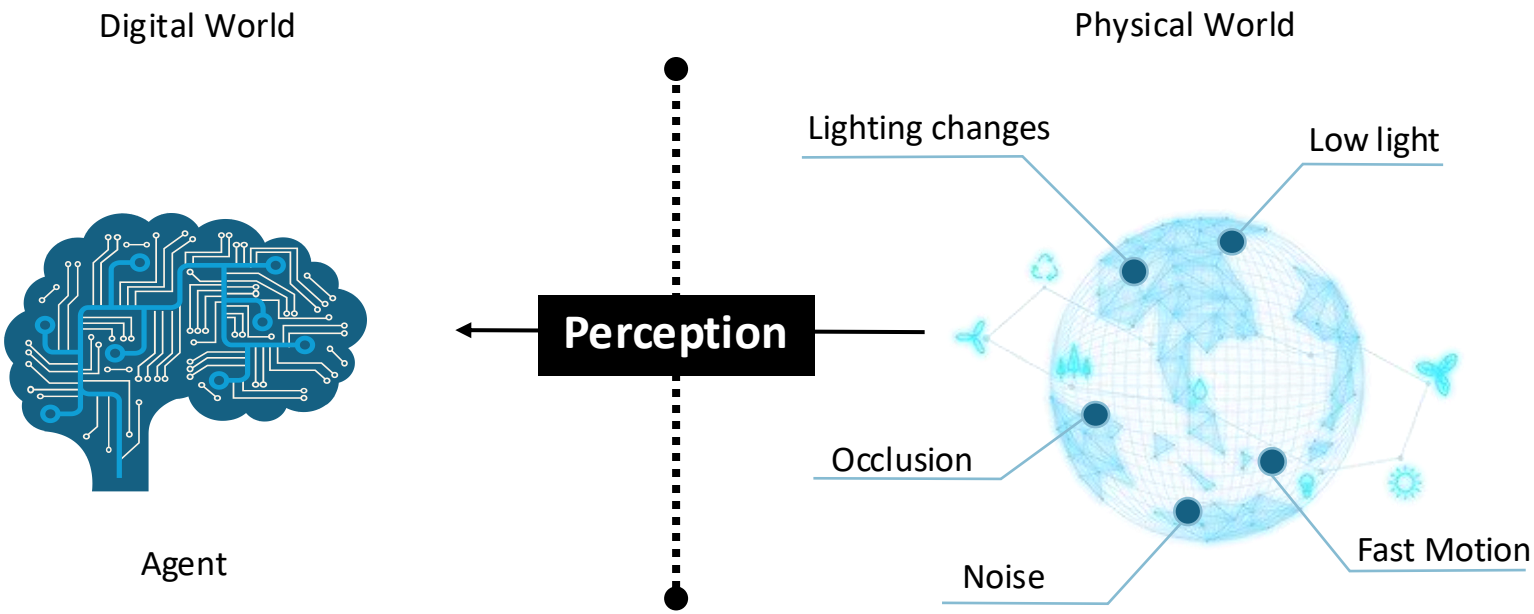
- Moravec, 1988, Mind Children

In Reality..

Building Robots to Perceive

https://www.youtube.com/watch?v=raHM3k-uR0E

Why is perception hard?



AI needs to **sense and understand** the unpredictable physical world

Objective of this course

- Establish a strong grounding in fundamental 3D geometry and computer vision
- Acquire core mathematical tools for multi-sensor perception (optimization, probability, estimation)
- Become proficient with state-of-the-art vision concepts and workflows
- Learn practical considerations in real robotic systems
- Industry job and research

Course Structure

- Module 1: Foundations of 3D Geometry
- Module 2: Localization and Reconstruction
- Module 3: Sensors in robotics
- Module 4: Modern perception for robotics

Grading

- 50 % assignments
- 1 Midterm, 20%
  - Tentatively 03/11 in class
- 30 % Group Final Project
  - Graded proposal + milestone
  - Presentations 05/04 and 05/06

Course material

- Slides posted after each class
- Canvas: announcement, post grades, Q&A (Discussion)
- Gradescope: upload assignments, code submission

Course format

- In-class lectures Hodson 311
- No zoom options will be provided

Assignments

- All assignments due at 11:59 pm
- Each assignment has two parts
  - Written questions (latex or scanned)
  - Coding (auto-graded)
- Each project is released when the previous project is due
- 5 free late days. No late submissions afterward.

Midterm

- Close-booked
- In-class written exam

Final Project

- Final project is an open-ended group project
- Groups will be assigned randomly
- Graded based on presentation and report
  - Two days for presentation
  - Report due last day of the semester
- Final project is in a publishable state

Policy Regarding AI Tools

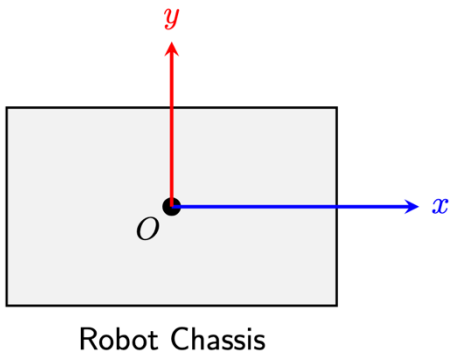
- You can use AI for grammar
- You cannot use AI for:
  - Writing code
  - Answering homework questions
- Copying other people's results (AI or not) is plagiarism

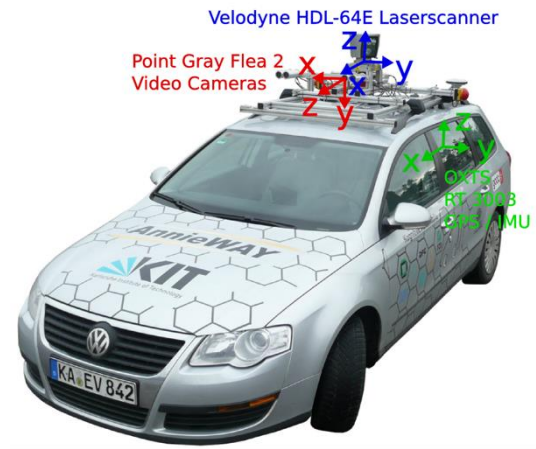
Compute Resources

- This course will not involve training big models
- GPUs will be main used for the last homework and final project
  - Student Colab is sufficient
- We will have a tutorial to help set up Colab and go over Pytorch basics
- Preferred if you have your own GPU for local testing

Coordinate Frames and Rigid Transformation

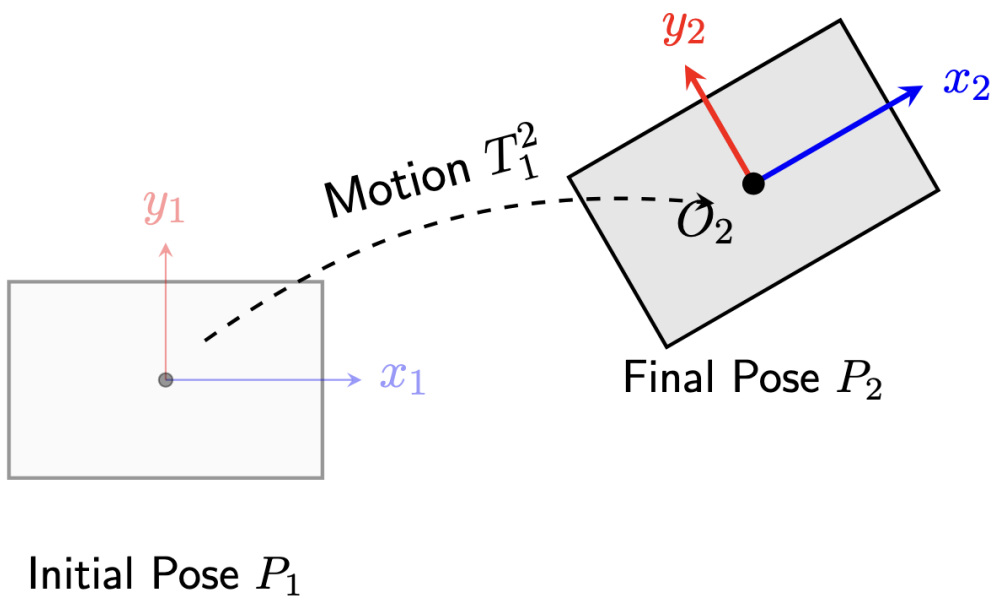
Pose of Robot



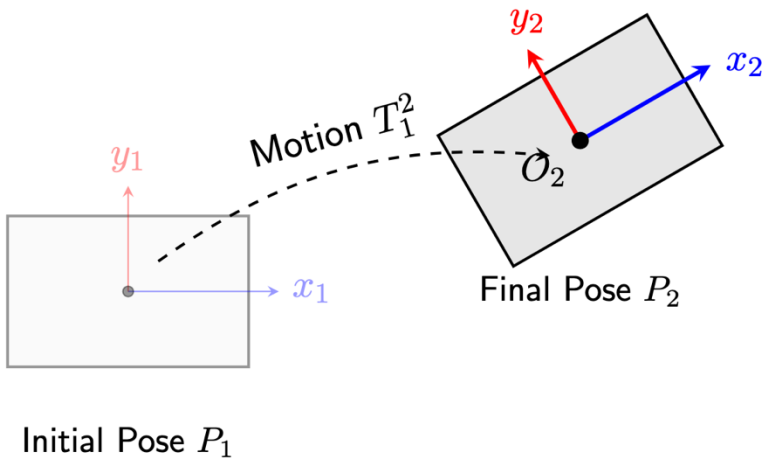


- The origin $(0, 0)$ is fixed at the robot's center of mass.

Rigid Transformation



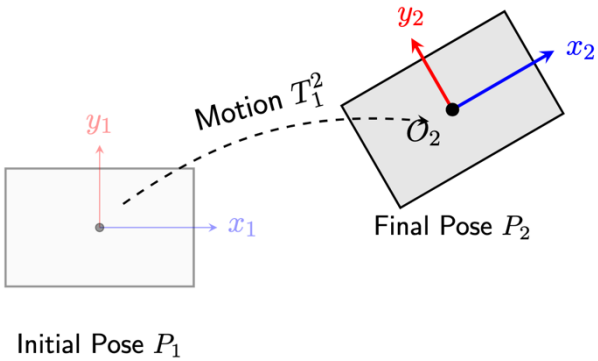
Rigid Transformation



Transformation Components:

- **Translation:** Moving the center from (x_1, y_1) to (x_2, y_2).
- **Rotation:** Changing the orientation by an angle θ.
- The motion is often represented by a transformation matrix $T \in SE(2)$.

Transform a frame



Original frame $\{1\}$:

$$O_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \hat{x}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \hat{y}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Transformation (in $\{1\}$): rotate by $45°$ and translate along the *original* axes

$$R = \begin{bmatrix} \cos 45° & -\sin 45° \\ \sin 45° & \cos 45° \end{bmatrix} = \frac{\sqrt{2}}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \qquad t = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ (m)}.$$

Effect on the frame:

$$\boxed{O_2 = RO_1 + t = t = \begin{bmatrix} 2 \\ 1 \end{bmatrix},}$$

points

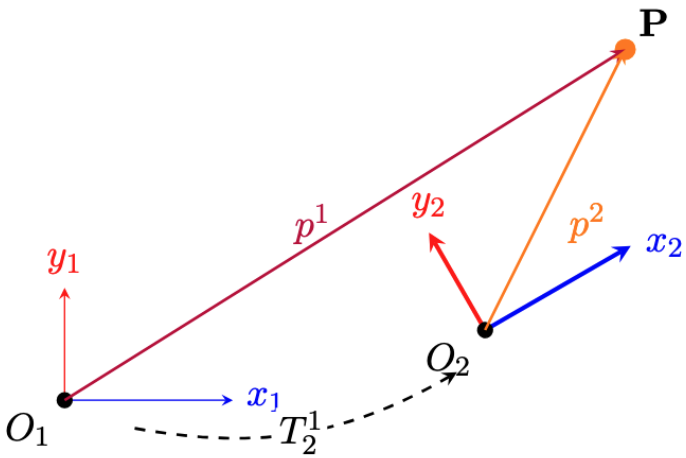
$$\boxed{\hat{x}_2 = R\hat{x}_1 = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}, \qquad \hat{y}_2 = R\hat{y}_1 = \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}.}$$

Vectors (only orientations)

Transform a point

Definitions (coordinates of the same physical point P)

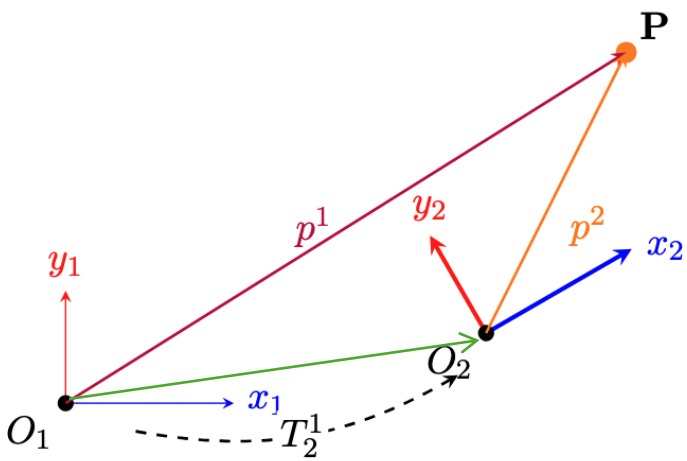
$$p^1 := {}^1\overrightarrow{O_1P}, \qquad p^2 := {}^2\overrightarrow{O_2P}, \qquad t_2^1 := {}^1\overrightarrow{O_1O_2}.$$



A vector does not have an origin!
A point = origin + vector

➔ **Viewing a vector in a different coordinate system is only a rotation**
➔ **No translation**

Transform a point



Definitions (coordinates of the same physical point P)

$$p^1 := {}^1\overrightarrow{O_1P}, \qquad p^2 := {}^2\overrightarrow{O_2P}, \qquad \boxed{t_2^1 := {}^1\overrightarrow{O_1O_2}.}$$

Vector identity

$$\overrightarrow{O_1P} = \overrightarrow{O_1O_2} + \overrightarrow{O_2P} \quad \Longrightarrow \quad \boxed{{}^1\overrightarrow{O_1P} = {}^1\overrightarrow{O_1O_2} + {}^1\overrightarrow{O_2P}.}$$

Rotate frame-2 coordinates into frame 1

Let R_2^1 map coordinates from frame 2 into frame 1. Then

$${}^1\overrightarrow{O_2P} = R_2^1\,{}^2\overrightarrow{O_2P} = R_2^1 p^2.$$

Result

$$\boxed{p^1 = t_2^1 + R_2^1 p^2.}$$

When adding vectors, make sure they are viewed in the same coordinate frame

2D Rotation

Definition (counterclockwise rotation by angle θ). For a vector $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2$, the rotated vector is

$$x' = R(\theta)\, x, \qquad R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

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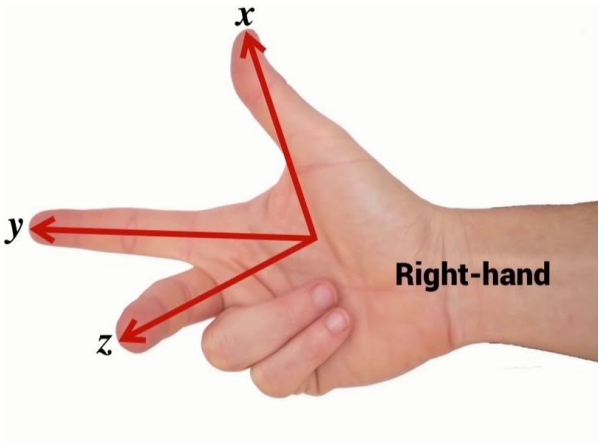
Expanded coordinate form.

$$\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \cos\theta - x_2 \sin\theta \\ x_1 \sin\theta + x_2 \cos\theta \end{bmatrix}.$$

3D Rotations



Definition (3D rotation). For a vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3, \qquad \mathbf{x}' = R\,\mathbf{x}, \quad R \in \mathbb{R}^{3\times 3}.$$

Common special case (rotation about a coordinate axis by angle θ).

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \quad R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \quad R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Check if a rotation is proper

$\boxed{R^\top R = I}$ (orthonormal columns/rows)

Let

$$R = \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 \end{bmatrix} \quad \text{(2D)}, \qquad R = \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 \end{bmatrix} \quad \text{(3D)}.$$

Check the dot products:

$$\mathbf{r}_i^\top \mathbf{r}_i = 1 \ \ \text{(unit length)}, \qquad \mathbf{r}_i^\top \mathbf{r}_j = 0 \ \ (i \neq j).$$

Equivalently,

$$R^\top R = I.$$

$\boxed{\det(R) = +1}$

-1 determinant means rotation + reflection
Converts a right-handed system to a left-handed one

Rotations form a group

A group is a set G with an operation $\circ$ satisfying:

- **Closure:** $a, b \in G \Rightarrow a \circ b \in G$
- **Associativity:** $(a \circ b) \circ c = a \circ (b \circ c)$
- **Identity:** $\exists e \in G$ s.t. $e \circ a = a \circ e = a$
- **Inverse:** $\forall a \in G,\ \exists a^{-1} \in G$ s.t. $a \circ a^{-1} = a^{-1} \circ a = e$

For planar rotations, take

$$\mathrm{SO}(2) = \{R(\theta) \in \mathbb{R}^{2\times 2} : R(\theta)^\top R(\theta) = I,\ \det(R(\theta)) = 1\}, \quad \circ \equiv \text{matrix multiplication.}$$

SO(3) Group

$$\mathrm{SO}(3) = \left\{ R \in \mathbb{R}^{3\times 3} \;\middle|\; R^\top R = I,\ \det(R) = 1 \right\}.$$

Elements of $\mathrm{SO}(3)$ represent **proper 3D rotations** (no reflection).

Key properties

- **Group operation:** matrix multiplication

$$R_1, R_2 \in \mathrm{SO}(3) \;\Rightarrow\; R_1 R_2 \in \mathrm{SO}(3).$$

- **Identity:** $I \in \mathrm{SO}(3)$.
- **Inverse:** $R^{-1} = R^\top \in \mathrm{SO}(3)$.
- **Non-commutative:** generally $R_1 R_2 \neq R_2 R_1$.

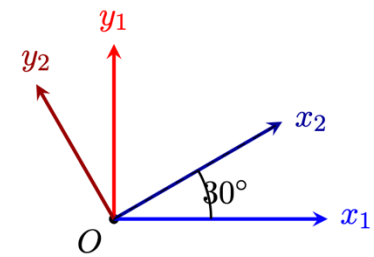
Examples of common groups and operations

- $(\mathbb{R}^n, +)$: vectors under addition (identity 0, inverse $-x$)
- $(\mathbb{R} \setminus \{0\}, \times)$: nonzero reals under multiplication (identity 1, inverse $1/x$)
- $\mathrm{SO}(2)$, $\mathrm{SO}(3)$ under matrix multiplication (rotations)
- $\mathrm{SE}(2)$, $\mathrm{SE}(3)$ under composition (rigid motions; can be represented by homogeneous matrices)

Example: how to read rotation matrices

R_2^1 maps coordinates from frame 2 into frame 1: $\mathbf{v}^1 = R_2^1 \mathbf{v}^2$.
A valid choice is the $30°$ CCW rotation matrix

$$R_2^1 = R(30°) = \begin{bmatrix} \cos 30° & -\sin 30° \\ \sin 30° & \cos 30° \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}.$$



Then each column is a unit axis direction:

$$R_2^1 = \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 \end{bmatrix}.$$

$\mathbf{r}_1 = {}^1\hat{\mathbf{x}}_2$ (x_2 axis expressed in frame 1), $\mathbf{r}_2 = {}^1\hat{\mathbf{y}}_2$ (y_2 axis expressed in frame 1).

Try yourself with this method: what is R_1^2?

$$R_1^2 = (R_2^1)^\top = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

Combining rotation and translation

Point (affected by translation)

$$\tilde{\mathbf{p}} = \begin{bmatrix} \mathbf{p} \\ 1 \end{bmatrix}, \qquad \mathbf{p} \in \mathbb{R}^n.$$

Rigid transform:

$$T = \begin{bmatrix} R & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix}, \qquad \tilde{\mathbf{p}}' = T\,\tilde{\mathbf{p}} = \begin{bmatrix} R\mathbf{p} + \mathbf{t} \\ 1 \end{bmatrix}.$$

Direction (not affected by translation)

$$\tilde{\mathbf{v}} = \begin{bmatrix} \mathbf{v} \\ 0 \end{bmatrix}, \qquad \mathbf{v} \in \mathbb{R}^n.$$

Same transform:

$$\tilde{\mathbf{v}}' = T\,\tilde{\mathbf{v}} = \begin{bmatrix} R\mathbf{v} \\ 0 \end{bmatrix}.$$

If $R \in \mathrm{SO}(3)$ (i.e., $R^\top R = I$ and $\det(R) = 1$), then $\boxed{T \in \mathrm{SE}(3).}$

Homogeneous Transformation

- Idea: represent points and rigid motions with a single matrix multiplication

point: $\tilde{\boldsymbol{p}} = \begin{bmatrix} \boldsymbol{p} \\ 1 \end{bmatrix}$

Rigid transform as one matrix

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ 0\,0\,0 & 1 \end{bmatrix}, \qquad \tilde{\boldsymbol{p}}' = \mathbf{T}\,\tilde{\boldsymbol{p}}$$

Composition becomes multiplication: $\mathbf{T}_A^C = \mathbf{T}_B^C \mathbf{T}_A^B$.

Composing Homogenous Transforms

- Depends on where the **change transform is defined**
- Can either be pre-multiplication or post-multiplication

Composition Order 1: Fixed (world)

Fixed frame (apply increment in the world frame)

Let T_B^W be the current pose of body frame B in world frame W. If an incremental motion ΔT is expressed in the *fixed* world frame, then

$$T_{B,\text{new}}^W = \Delta T \, T_B^W \qquad \text{(pre-multiply).}$$

Composition Order 2: Changing Body

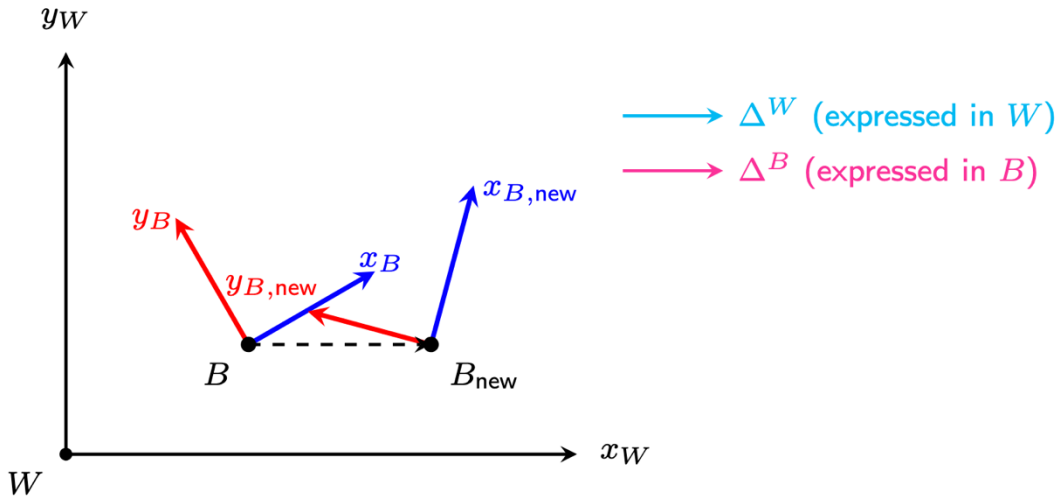
Changing frame (apply increment in the body frame)

If the incremental motion ΔT is expressed in the *current body frame* B, then

$$T^W_{B,\text{new}} = T^W_B \, \Delta T \qquad \text{(post-multiply).}$$

Example – Composition Order

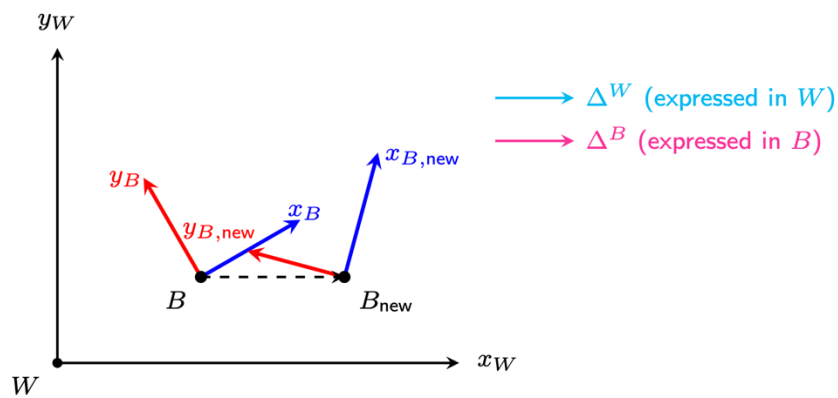
Key idea: Same effective pose change $T^W_{B,\text{new}}$; different increment representations Δ^W vs Δ^B.



Translation B->B_new is different in two frames

Example – Composition Order

Key idea: Same effective pose change $T^W_{B,\text{new}}$; different increment representations Δ^W vs Δ^B.



World increment (expressed in W)

$$T^W_{B,\text{new}} = \Delta^W T^W_B$$

$$\Delta^W := T^W_{B,\text{new}} (T^W_B)^{-1}$$

Body increment (expressed in B)

$$T^W_{B,\text{new}} = T^W_B \Delta^B$$

$$\Delta^B := (T^W_B)^{-1} T^W_{B,\text{new}}$$

Rule of thumb: fixed frame $\Rightarrow$ pre-multiply; changing frame $\Rightarrow$ post-multiply.

Example: chaining multiple transforms

Notation. Let $\mathbf{T}_Y^X \in SE(3)$ map coordinates *from frame* Y *to frame* X:

$$\mathbf{p}^X = \mathbf{T}_Y^X \, \mathbf{p}^Y, \qquad \mathbf{T}_Y^X = \begin{bmatrix} \mathbf{R}_Y^X & \mathbf{t}_Y^X \\ \mathbf{0}^\top & 1 \end{bmatrix}.$$

Camera to World through A.

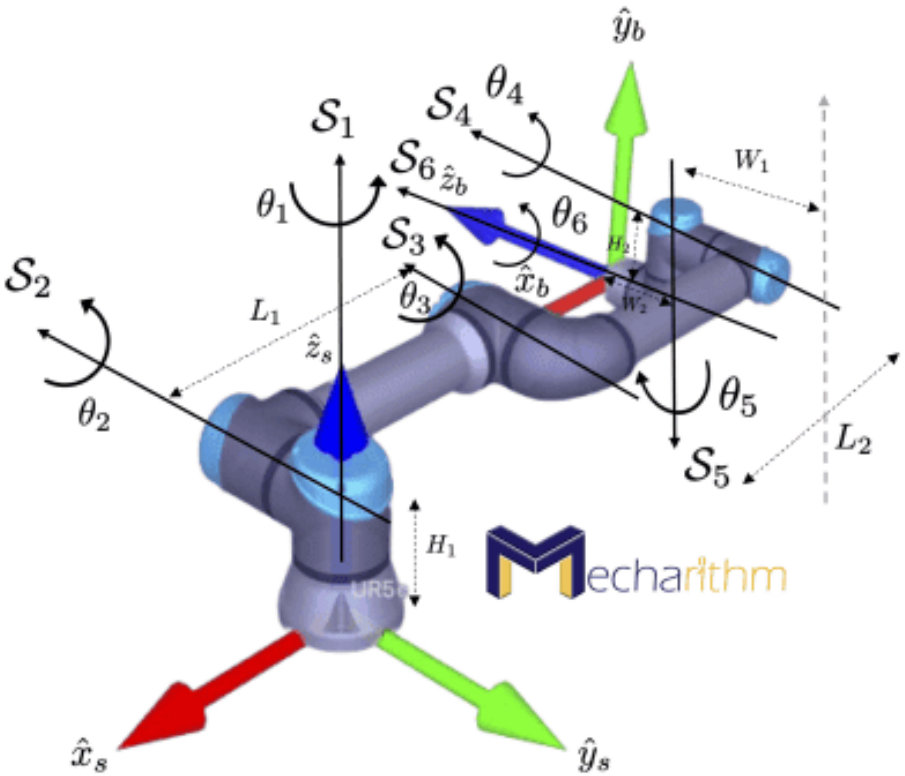
$$\mathbf{p}^W = \mathbf{T}_A^W \, \mathbf{T}_C^A \, \mathbf{p}^C, \qquad \Rightarrow \mathbf{T}_C^W = \mathbf{T}_A^W \, \mathbf{T}_C^A.$$

"Cancellation" intuition (frame labels).

$$\mathbf{p}^W = \underbrace{\mathbf{T}^W{}_A}_{\text{takes } A \to W} \; \underbrace{\mathbf{T}^A{}_C}_{\text{takes } C \to A} \; \mathbf{p}^C \implies \mathbf{T}^W{}_C.$$

The adjacent *output* frame of the right transform (A) matches the adjacent *input* frame of the left transform (A), so the intermediate frame is internal to the chain.

Forward kinematics: Post- or pre-multiplication?



Parameterization of Rotations

Common Parameterizations

- Rotations Matrices (9 values)
- Axis Angle (3 values)
- Quaternion (4 values)
- Euler Angles (3 * N values)

1. Axis-Angle

Axis–angle:

$$\boldsymbol{u} \in \mathbb{R}^3, \quad \|\boldsymbol{u}\| = 1, \quad \theta \in [0, \pi]$$

Exponential coordinates:

$$\boxed{\boldsymbol{\phi} = \theta \boldsymbol{u}}$$

Also called "rotation vector"

Conversions:

$$\boldsymbol{u} = \frac{\boldsymbol{\phi}}{\|\boldsymbol{\phi}\|}, \qquad \theta = \|\boldsymbol{\phi}\|$$

Interpretation: exponential coordinates combine axis and magnitude into a single vector.

Question: is this unique?

1. Axis-Angle to rotation matrix

Any rotation can be written as an exponential of a skew-symmetric matrix:

$$R = \exp(\phi^\wedge), \qquad \phi = \theta\, u, \quad \|u\|_2 = 1,$$

where u is the rotation axis and θ is the rotation angle.

Rodrigues' formula

$$R = I + \sin\theta\, u^\wedge + (1 - \cos\theta)\,(u^\wedge)^2.$$

where

$$\omega^\wedge := \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

Example: Rodrigue's Formula

$$\boldsymbol{\omega} = \begin{bmatrix} 0 \\ 0 \\ \pi/6 \end{bmatrix} = \theta\,\mathbf{u}, \qquad \theta = \|\boldsymbol{\omega}\| = \pi/6, \quad \mathbf{u} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

$$[\mathbf{u}]_\times = \begin{bmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad [\mathbf{u}]_\times^2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$R = I + \sin\theta\,[\mathbf{u}]_\times + (1-\cos\theta)\,[\mathbf{u}]_\times^2.$$

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \sin(\pi/6) \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \big(1-\cos(\pi/6)\big) \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$R = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Axis-Angle from rotation matrix

Goal. Given $R \in \mathrm{SO}(3)$, compute the rotation vector $\phi \in \mathbb{R}^3$ such that

$$R = \exp(\phi^\wedge), \qquad \phi = \theta\, u.$$

Step 1: angle from the trace.

$$\theta = \cos^{-1}\left(\frac{\mathrm{tr}(R) - 1}{2}\right) \in [0, \pi].$$

Step 2: axis from the skew part (generic case $\theta \not\approx 0, \pi$).

$$u^\wedge = \frac{1}{2\sin\theta}\,(R - R^\top), \qquad u = (u^\wedge)^\vee, \qquad \phi = \theta\, u.$$

Example: axis-angle from rotation matrix

$$R = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 1 (Angle from trace).

$$\theta = \arccos\left(\frac{\mathrm{tr}(R) - 1}{2}\right), \qquad \mathrm{tr}(R) = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} + 1 = \sqrt{3} + 1.$$

$$\Rightarrow \quad \cos\theta = \frac{\mathrm{tr}(R) - 1}{2} = \frac{\sqrt{3}}{2} \quad \Longrightarrow \quad \theta = \arccos\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}.$$

Step 2 (Axis from skew-symmetric part).

$$\mathbf{a} = \begin{bmatrix} R_{32} - R_{23} \\ R_{13} - R_{31} \\ R_{21} - R_{12} \end{bmatrix} = \begin{bmatrix} 0 - 0 \\ 0 - 0 \\ \frac{1}{2} - (-\frac{1}{2}) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

$$\mathbf{u} = \frac{1}{2\sin\theta}\,\mathbf{a} = \frac{1}{2\sin(\pi/6)} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Step 3 (Rotation vector).

$$\boldsymbol{\omega} = \theta\,\mathbf{u} = \frac{\pi}{6} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \frac{\pi}{6} \end{bmatrix}.$$

Compute angular velocity

Given two orientations $R_1, R_2 \in \mathrm{SO}(3)$ (e.g., at times t_1 and $t_2 = t_1 + \Delta t$).

Relative rotation. The rotation taking frame-1 orientation to frame-2 orientation is

$$R_{21} := R_2\, R_1^\top \in \mathrm{SO}(3).$$

Axis–angle via the log map. Define the rotation vector

$$\phi := \mathrm{Log}(R_{21})^\vee \in \mathbb{R}^3, \qquad \theta := \|\phi\|_2, \qquad u := \begin{cases} \phi/\theta, & \theta \neq 0 \\ \text{any unit axis}, & \theta = 0 \end{cases}$$

so that $R_{21} = \exp(\phi^\wedge) = \exp(\theta\, u^\wedge)$.

Angular velocity (approx. constant over Δt).

$$\omega_{\text{world}} \approx \frac{1}{\Delta t}\,\phi \qquad \text{(expressed in world frame when } R_{21} = R_2 R_1^\top\text{)}$$

*Alternative approach for angular velocity

Continuous time (time-varying R)

Assume $R(t) = R_B^W(t) \in \mathrm{SO}(3)$ maps coordinates from the body frame B to the world frame W:

$$v^W = R_B^W \, v^B.$$

Its time derivative satisfies $\dot{R}\, R^\top \in \mathfrak{so}(3)$.

World-frame angular velocity

$$\hat{\omega}^W := \dot{R}\, R^\top \in \mathfrak{so}(3), \qquad \omega^W := (\hat{\omega}^W)^\vee \in \mathbb{R}^3.$$

Body-frame angular velocity

$$\hat{\omega}^B := R^\top \dot{R} \in \mathfrak{so}(3), \qquad \omega^B := (\hat{\omega}^B)^\vee \in \mathbb{R}^3.$$

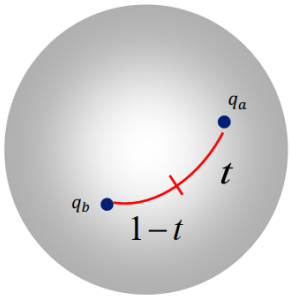
Relationship

$$\hat{\omega}^W = R\,\hat{\omega}^B\, R^\top, \qquad \omega^W = R\,\omega^B, \qquad \omega^B = R^\top \omega^W.$$

Rotations are special

- Rotations cannot be simply added together
- Computing difference and velocity is not as simple as translation
- Averaging and interpolating need special methods
  - Geodesic midpoint
  - Slerp interpolation with quaternions

$$R_{\text{avg}} = R_1 \exp\left(\frac{1}{2} \log(R_1^\top R_2)\right)$$



- $\mathbf{q_t} = \frac{\sin((1-t)\theta)}{\sin\theta}\mathbf{q_a} + \frac{\sin(t\theta)}{\sin\theta}\mathbf{q_b}$
- $\theta = \mathrm{acos}(\mathbf{q_a} \cdot \mathbf{q_b})$

Source: https://starrykss.tistory.com/1806

Robots have many frames

```yaml
cam0:
  cam_overlaps: [1, 2, 3, 4]
  camera_model: pinhole
  distortion_coeffs: [-0.04942202185899467, 0.07765947332434518, 0.000877334696786852, -0.0015249203609635062]
  distortion_model: radtan
  intrinsics: [1027.7306775414688, 1027.2217271273582, 631.5864157958096, 368.42668838653725]
  resolution: [1280, 720]
  rostopic: /prophesee/left/events
cam1:
  T_cn_cnm1:
  - [0.9999826194000789, -0.0008369998397056543, -0.005836122773548713, -0.1201967905613849]
  - [0.0008245516921100342, 0.9999973808811554, -0.0021350281822576024, 0.001000917907982953]
  - [0.005837894506295886, 0.002130178889278652, 0.9999806904763889, 0.0002487790410553178]
  - [0.0, 0.0, 0.0, 1.0]
  cam_overlaps: [0, 2, 3, 4]
  camera_model: pinhole
  distortion_coeffs: [-0.056059056656929326, 0.08282758641317992, 0.0018278285190087514, -3.057854809916077e-05]
  distortion_model: radtan
  intrinsics: [1027.6397033251828, 1027.515094529007, 634.5684955427264, 370.1640336417698]
  resolution: [1280, 720]
  rostopic: /prophesee/right/events
cam2:
  T_cn_cnm1:
  - [0.999916184055119, -0.010025433046204221, 0.008192408490601974, 0.12026585333591912]
  - [0.010003843538660403, 0.9999463901937963, 0.0026720503066773337, 0.06909336942544439]
  - [-0.00821875775861617, -0.002589870773511346, 0.9999628716058777, 0.0028350077406005584]
  - [0.0, 0.0, 0.0, 1.0]
  cam_overlaps: [0, 1, 3, 4]
  camera_model: pinhole
  distortion_coeffs: [-0.39869675623484474, 0.16445165646978754, -0.00013071600753056993, 0.0015013956534575509]
  distortion_model: radtan
  intrinsics: [1053.897816983714, 1053.2900731843688, 669.2007802033011, 335.62603230615827]
  resolution: [1280, 800]
  rostopic: /ovc/left/image_mono/compressed
cam3:
  T_cn_cnm1:
  - [0.9999977733329843, 0.0004901817997415576, 0.002052571771028786, -0.11999476789321858]
  - [-0.0004911607677823989, 0.9999997658686255, 0.0004764701386342235, -5.9418783413758316e-05]
  - [-0.0020523377334672583, -0.00047747722042086913, 0.9999977799602009, 0.0011937541060889087]
  - [0.0, 0.0, 0.0, 1.0]
  cam_overlaps: [0, 1, 2, 4]
```

Takeaways

- How do we represent a robot with a set of transforms?
- How do we move a rigid body (rotation + translation)?
- Read and examine a rotation matrix
- Conversion between a rotation vector and a rotation matrix
- Compute angular velocity between two transforms